

CodeAlpha (DevOps) task : 01

TITLE: Jenkins Remoting Project

AIM: To configure Jenkins Remoting for connecting a Jenkins Agent to a Jenkins Controller, execute jobs remotely, distribute build workloads, and demonstrate secure remote execution using Jenkins.

OBJECTIVES

1. Set up Jenkins Remoting to connect remote Jenkins nodes.
2. Distribute build loads across machines securely.
3. Execute jobs remotely on Jenkins Agents.
4. Implement node isolation using labels.
5. Gain hands-on experience with Jenkins remote execution capabilities.

SOFTWARE REQUIREMENTS

- Windows 11
- Java JDK 25
- Jenkins LTS
- Web Browser
- Command Prompt

THEORY

Jenkins is an open-source automation server used to implement Continuous Integration (CI) and Continuous Deployment (CD). Jenkins Remoting is the communication mechanism that enables a Jenkins Controller to communicate with one or more Jenkins Agents.

A Controller manages and schedules jobs, while Agents execute the assigned jobs. This architecture helps distribute workloads, improve performance, and support multiple environments.

In this project, Jenkins Remoting is configured using agent.jar and WebSocket communication. The Agent authenticates itself using a secret key generated by Jenkins and executes jobs remotely.

PROCEDURE

Step 1: Install Java

1. Download and install Java JDK.
2. Verify installation using:

Command:

```
java -version
```

Output:

```
java version "25.0.1"
```

Step 2: Install Jenkins

1. Download Jenkins LTS version.
2. Install Jenkins using Jenkins MSI installer.
3. Configure port 8080.
4. Start Jenkins Service.
5. Open Jenkins using:

<http://localhost:8080>

6. Unlock Jenkins using the initial admin password.

Step 3: Install Plugins

1. Select "Install Suggested Plugins".
2. Wait for plugin installation.

3. Create Administrator Account.

Step 4: Create Jenkins Agent

1. Navigate to:

Manage Jenkins → Nodes

2. Click:

New Node

3. Enter:

Node Name: Agent1

4. Select:

Permanent Agent

5. Configure:

Number of Executors: 2

Remote Root Directory:

C:\JenkinsAgent

Label:

windows

Launch Method:

Launch agent by connecting it to the controller

6. Save the configuration.

Step 5: Create Agent Directory

Create directory:

C:\JenkinsAgent

Step 6: Connect Agent Using Jenkins Remoting

Open Agent1 page and copy generated command.

Command:

```
java -jar agent.jar -url http://localhost:8080/ -secret <generated_secret> -name Agent1 -webSocket  
-workDir "C:\JenkinsAgent"
```

Expected Output:

INFO: WebSocket connection open

INFO: Connected

This confirms successful Agent connection.

Step 7: Create Build Job

1. Click New Item.
2. Enter:

Remote-Test

3. Select:

Freestyle Project

4. Click OK.

Step 8: Configure Remote Execution

Enable:

Restrict where this project can be run

Label:

windows

This ensures the job runs only on Agent1.

Step 9: Add Build Steps

Build Step:

Execute Windows Batch Command

Commands Used:

echo Running on Jenkins Agent

hostname

java -version

echo Build Successful

Step 10: Execute Build

Click:

Build Now

Jenkins dispatches the job to Agent1 through Jenkins Remoting.

Step 11: Verify Console Output

Observed Output:

Building remotely on Agent1 (windows)

Running on Jenkins Agent

hp

```
java version "25.0.1"
```

```
Build Successful
```

```
Finished: SUCCESS
```

The output confirms successful remote execution on Agent1.

COMMANDS USED

1. Verify Java

```
java -version
```

2. Navigate to Agent Directory

```
cd C:\JenkinsAgent
```

3. Connect Agent

```
java -jar agent.jar -url http://localhost:8080/ -secret <generated_secret> -name Agent1 -webSocket  
-workDir "C:\JenkinsAgent"
```

4. Build Commands

```
echo Running on Jenkins Agent
```

```
hostname
```

```
java -version
```

```
echo Build Successful
```

OBSERVATIONS

- Jenkins Controller installed successfully.
- Agent1 configured successfully.

- WebSocket connection established.
- Agent authenticated using secret key.
- Remote job executed successfully.
- Build output received from Agent1.

RESULT

Jenkins Remoting was successfully configured by connecting Agent1 to the Jenkins Controller using agent.jar and WebSocket communication. A remote build job was executed successfully on Agent1. The project demonstrated remote execution, node isolation, workload distribution, and secure communication between Controller and Agent.

CONCLUSION

The experiment successfully demonstrated Jenkins Remoting. A Jenkins Agent was connected to the Jenkins Controller and used to execute build jobs remotely. The implementation provided practical knowledge of distributed build execution, secure agent communication, and Jenkins node management.

Screenshots which shows that the project has been successfully perform:

First screenshot:

```
INFO: Using C:\JenkinsAgent\remoting as a remoting work directory
Jun 23, 2026 9:01:54 PM org.jenkinsci.remoting.engine.WorkDirManager setupLogging
INFO: Both error and output logs will be printed to C:\JenkinsAgent\remoting
Jun 23, 2026 9:01:55 PM hudson.remoting.Launcher createEngine
INFO: Setting up agent: Agent1
Jun 23, 2026 9:01:55 PM hudson.remoting.Engine startEngine
INFO: Using Remoting version: 3355.v388858a_47b_33
Jun 23, 2026 9:01:55 PM org.jenkinsci.remoting.engine.WorkDirManager initializeWorkDir
INFO: Using C:\JenkinsAgent\remoting as a remoting work directory
Jun 23, 2026 9:01:55 PM hudson.remoting.Launcher$CuiListener status
INFO: WebSocket connection open
Jun 23, 2026 9:01:55 PM hudson.remoting.Launcher$CuiListener status
INFO: Connected
```

Second screenshot:

```
C:\JenkinsAgent\workspace\Remote-Test>java -version
java version "25.0.1" 2025-10-21 LTS
Java(TM) SE Runtime Environment (build 25.0.1+8-LTS-27)
Java HotSpot(TM) 64-Bit Server VM (build 25.0.1+8-LTS-27, mixed mode, sharing)

C:\JenkinsAgent\workspace\Remote-Test>echo Build Successful
Build Successful

C:\JenkinsAgent\workspace\Remote-Test>exit 0
Finished: SUCCESS
```

Third screenshot:

Nodes

+ New Node

Configure Monitors

Add, remove, control and monitor the various nodes that Jenkins runs jobs on.

S	Name ↓	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	Agent1	Windows 11 (amd64)	In sync	343.27 GiB	4.78 GiB	343.27 GiB	95ms
	Built-In Node	Windows 11 (amd64)	In sync	343.27 GiB	4.79 GiB	343.27 GiB	0ms
Data obtained		12 min	12 min	12 min	12 min	12 min	12 min

Icon: S M **L**

Legend